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# A multi-radar emitter sorting and recognition method based on hierarchical clustering and TFCN

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ABSTRACT

Radar emitter recognition is an important part of the electronic attack and defense system, and its key task is to sort and identify various information of emitters from mixed pulse streams. Pulse descriptor word, as an easily obtainable feature, is often used for sorting and recognition. Aiming at the problems of high complexity and difficulty in handling intra-class clustering and inter-class aliasing of existing sorting methods, a hierarchical clustering method based on Kernel Density Estimation-Kullback-Leibler Divergence-Template Matching (KDE-KLD-TM) is proposed. Firstly, the down-sampled data is used to construct central clusters, greatly improving the processing speed. Then, with probability theory as the theoretical support, inter-class clustering on all samples is performed based on maximum posterior probability. Finally, based on cluster distribution similarity and periodic template matching, intra-class merging and inter-class deinterleaving are completed. After sorting, considering the periodic differences in pulse repetition intervals among different types of emitter and the insufficient attention paid by existing recognition methods to this feature, a time-frequency convolution network (TFCN) based emitter recognition method is proposed for the first time in terms of pulse description word. Using time-frequency analysis (TFA) to extract periodic features and using convolutional neural networks (CNN) for classification, the one-dimensional sequence classification problem is treated as a two-dimensional image classification problem. The proposed method is simulated in a typical scenario with intra-class clustering and inter-class aliasing. The results show that the proposed method can sort a total of 2.06 million aliased samples composed of 10 classes within 58.61 s, and the recognition accuracy reaches 96.33%. The comparison with the baseline method proves the effectiveness and progressiveness of the proposed method. Finally, the sorting and recognition performance of the proposed algorithm in complex scenarios in the presence of pulse loss is discussed.

## 1. Introduction

Radar emitter recognition technology preprocesses electromagnetic signals intercepted by a receiver, segregates radar signals, extracts multi-domain features from these refined radar signals, and designs a classifier to obtain the recognition results [1]. Compared to traditional signal type recognition, the results of radar emitter recognition can include signal modulation types [2], radar operating modes [3], radar types [4], and even specific radar individual [5]. A stable and reliable recognition technology is a prerequisite for effective countermeasures in electronic warfare systems, making radar emitter recognition a persistent research focus in the field of electronic warfare [6].

The Pulse Descriptor Word (PDW) is used to characterize inter-pulse parameter features of radar signals, including the radio frequency (RF), pulse width (PW), pulse amplitude (PA), time of arrival (TOA), and direction of arrival (DOA) [7]. These features can be directly measured through signal detection, and the methods for obtaining them are highly developed and widely applied in numerous equipment systems. They are commonly used as sorting and recognition features [8].

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Successful sorting is a prerequisite for accurate emitters recognition [9]. Especially in complex electromagnetic environments where multiple categories of signals are aliasing, PDWs with adjacent arrival times may belong to different emitters, making it difficult to extract statistical and multi-domain processing features from the PDWs. This severely limits the available feature extraction techniques and recognition methods. Clustering is a common method for sorting emitters, which can group similar samples according to specific rules, thereby enabling the extraction and concatenation of single-class temporal samples [10]. Note that clustering is an unsupervised algorithm that does not provide label information in its output. Therefore, it must be combined with a supervised classifier for emitter recognition. Common clustering algorithms include K-means [11], Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [12], and various improved versions thereof [13,14]. K-means requires the number of clusters to be specified, which can be challenging due to the lack of prior information needed for practical scenarios. DBSCAN, a density-based clustering method, does not require the number of clusters to be predefined but is sensitive to the setting of neighborhood radius. The main issues with these clustering methods are high computational complexity—K-means involves iterative refinement, while DBSCAN has complexity on the order of the square of the total number of samples—and a high likelihood of splitting a single class into multiple classes or merging multiple classes into one, both of which can negatively impact subsequent recognition. To address the aforementioned challenges, hierarchical clustering methods represent a viable solution [15]. Nonetheless, existing approaches demand repeated iterations across the entire dataset, leading to substantial computational overhead and high time complexity. These limitations render them less suitable for large-scale data processing tasks. Furthermore, the irreversibility of decisions regarding cluster merging or splitting increases the risk of generating inaccurate clustering results.

With the rapid development of artificial intelligence, recognition methods based on PDWs have evolved from template matching [16] to machine learning classifiers [17], and more recently to deep learning classifiers. Deep learning can further extract deep features from raw data or manually extracted features using neural networks for classification purposes [18]. In [19], the first-order differences of time-of-arrival (DTOA), i.e., pulse repetition intervals (PRI), along with RF and PW, were used to construct a three-dimensional feature image. Based on a convolutional neural network (CNN), this approach achieved the recognition of 58 types of radar emitters. In [20], an attempt was made to map RF, PW, DOA, and PRI onto a polar coordinate system and use CenterNet to detect the type and location of emitters. In [21], DBSCAN was applied to sort emitters based on RF, PW, PA, and DOA. Then, RF and PRI were used as recognition features, and the effectiveness of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks in identifying radar operating modes was discussed.

Time-frequency analysis (TFA) is a common technique for analyzing non-stationary signals and has been proven by some scholars to offer significant advantages when combined with CNNs in the field of emitter recognition [22,23]. Current research in this area mainly focuses on two aspects. The first involves denoising the time-frequency images to improve recognition performance under low signal-to-noise ratios (SNRs) [24,25]. The second aspect revolves around studying different TFA methods and neural network designs. For instance, in [26–28], short-time Fourier transform (STFT), Choi-William distribution (CWD), and Wigner-Ville distribution (WVD) were used to obtain time-frequency images, which were then fed into CNNs for signal modulation type recognition. In terms of network architecture design, [29] proposed a CNN capable of processing multiple time-frequency images, and used a fully connected structure to integrate the results, improving recognition performance. [30] utilized a squeeze-and-excitation network (SENet) for signal modulation type recognition, integrating the discriminative results from the time, frequency, and time-frequency domains as the final output. [31] fed both the time-frequency images obtained from CWD and statistical features of these images into a CNN and an Elman neural network (ENN) respectively, achieving the recognition of 12 modulation types. However, all of the above studies focus on applying TFA to the fast-time dimension to recognize intra-pulse modulation features, and there has been no discussion on the combination of TFA and inter-pulse features such as PDW.

In response to the aforementioned issues and current research trends, this paper proposes a method for multi-radar emitter sorting and recognition based on hierarchical clustering and time-frequency convolution network (TFCN). The dataset used in this study consists of PDW features and their labels provided by authoritative research institutions, but the meaning of the labels is unspecified. Therefore, we do not specify the exact form of the emitter recognition results but abstract them into general classes. Through feature consistency and separability analysis of the PDW dataset, we selected DOA and RF as clustering features and PRI as a recognition feature, laying the foundation for subsequent algorithm development. For sorting, we propose a hierarchical clustering method based on Kernel Density Estimation-Kullback-Leibler Divergence-Template Matching (KDE-KLD-TM). First, we down-sample the entire sample set to obtain central clusters, significantly reducing computational complexity. Then, we compute the maximum posterior probability of each sample relative to each central clusters to perform clustering, with theoretical guarantees for clustering performance based on probability theory. Next, we calculate the KL divergence between clusters and merge similar ones according to distribution similarity. Finally, based on the changes in density, we judge aliasing segments and pure segments within each cluster, and extract periodic templates from the pure segments to purify the aliasing segments. For identification, we propose an emitter recognition method based on TFCN. We first arrange PRI along the slow time dimension as a time series rather than treating them as independent samples, which enhances feature separability by introducing temporal relationships. Then, we apply STFT to the time series to extract the time-frequency features of PRI, and use EfficientNet to recognize the classes of radar emitters.

The main contributions of this paper are summarized as follows:

1) To address the issues of slow speed and difficulty in handling intra-class clustering and inter-class aliasing in existing clustering methods, we propose a hierarchical clustering method based on KDE-KLD-TM. This method significantly reduces the interaction between samples, has solid theoretical support, and effectively solves the problems of merging clusters of the same class and deinterleaving between different classes.

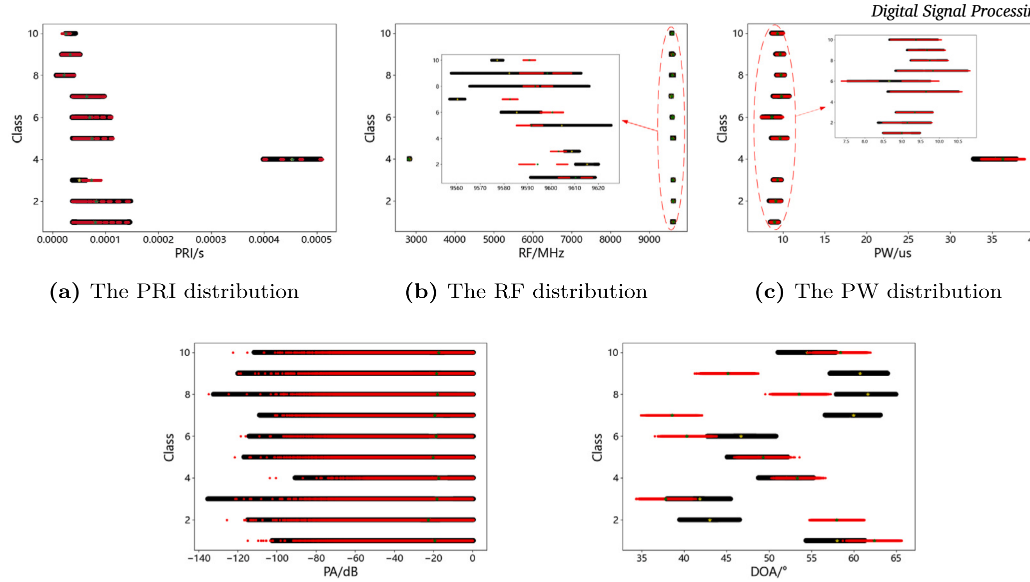
2) To address the problem of high inter-class feature distribution overlap and poor separability in original PDW features, we propose an emitter recognition method based on TFCN for the first time in the context of PDW. By arranging discrete PRI samples into a slow-time sequence and extracting periodic features between different classes using TFA along the slow time dimension, we leverage the differences in periodicity for recognition.

The structure of this paper is as follows. In Section 2, the external dataset used for training and testing is introduced and the challenges faced by PDW features in sorting and recognition are analyzed. In Section 3, the proposed hierarchical clustering and TFCN method for emitter sorting and recognition is presented. In Section 4, simulation results and analysis of the proposed method are provided. Finally, conclusions are drawn in Section 5.

## 2. Data description and feature analysis

This section introduces a simulated dataset used for subsequent algorithm research, which includes two typical challenges: intra-class feature distributions with clustering and inter-class feature distributions with overlapping. These data were constructed by experts in this field, and we do not have any parameter information about the transmitter and receiver. Although anonymous, it is labeled to ensure that we can conduct supervised training and algorithm performance testing.

The dataset encompasses 10 emitter classes, and each sample is described by a 5-dimensional PDW feature of $\{\mathrm{TOA}, \mathrm{RF}, \mathrm{PW}, \mathrm{PA}, \mathrm{DOA}\}$,




**(a)** The PRI distribution **(b)** The RF distribution **(c)** The PW distribution

**(d)** The PA distribution **(e)** The DOA distribution

**Fig. 1.** The 5 characteristic distributions of 10 emitter classes.

measured in units of seconds (s), megahertz (MHz), microseconds (μs), decibels (dB), and degrees (°) respectively. The values of these features vary within certain ranges but do not contain noise. Our research is based solely on these aforementioned features, without considering intra-pulse modulation information.

The dataset is composed of a train set and a test set. In the train set, samples of each emitter class are collected independently, without aliasing, loss, error, or interference. Samples of each emitter are sorted by their time of arrival, with a total collection duration of 10 seconds per class. The number of samples per class is approximately between 120,000 to 220,000, with a total of about 1.4 million samples. The test set sorts samples from the 10 emitter classes based on different times of arrival, collecting for a total of 16 seconds. The number of samples per class is roughly between 180,000 to 350,000, with a total of about 2.06 million samples. Note that each TOA corresponds to only one emitter class, thus one class label.

Fig. 1 shows scatter plots of five feature distribution ranges for both the train and test sets, where black and red points represent the train and test samples respectively, and yellow and green points correspond to their respective means. Since the TOA merely reflects the temporal sequence of samples, it is common practice to use the first-order difference of TOA, known as the PRI, as a discriminative feature [32,33]. Therefore, this paper chooses PRI instead of TOA as the feature for analysis.

This paper employs two metrics, separability and consistency, to qualitatively evaluate the distribution range of the aforementioned features. Separability indicates the degree of overlap in the distribution ranges of different emitter classes under a certain feature, whereas consistency measures the degree of alignment in the distribution range of a given feature for the same emitter class across both the train and test sets. We aim to select features with good separability and consistency for subsequent sorting and recognition.

From Fig. 1, it is visually apparent that for the above five features, except for Class 4, the features of various emitter classes exhibit significant overlaps in both the train and test sets, making it challenging to distinguish any single emitter class from all others. However, DOA and RF can achieve separation of at least two classes simultaneously in both the train and test sets, with local separability. Therefore, these two features are selected for sorting. Notably, RF shows intra-class clustering in Classes 1, 2, and 9 of the test set. If the sorting method fails to merge these clusters, it may subsequently affect the recognition of these classes.

Considering consistency, PRI, PA, and PW are features with better consistency, whereas RF and DOA show noticeable inconsistencies in distribution between the train and test sets. Considering the generalization performance of classifiers, avoiding concept drift between different scenarios is crucial, making consistency the primary criterion for selecting classification features. Although PRI, PA, and PW demonstrate good consistency, their separability is poor, meaning that the value ranges of different emitter classes under these features are quite similar, thus making it difficult to identify based on a single sample alone. Inspired by the coupled relationship between different operating modes and signal parameter variations in multifunctional radars, multiple samples of a certain class can be regarded as periodic sequential data. The disparities in feature values as they change over time can aid in the process of recognition [34]. By comparing the periodic consistency, PRI is better than PA and PW, so PRI is chosen as the feature for recognition.

Fig. 2 shows the periodic sequences $\left\{\mathbf{S}_i\right\}_{i=1}^{10}$ obtained by folding all samples of the 10 classes according to the PRI cycle, where $\mathbf{S}_i = \left\{\mathbf{s}_{i,1}, \mathbf{s}_{i,2}, \ldots, \mathbf{s}_{i,q_i}\right\} \in \mathbb{R}^{T_i \times q_i}$ represents the $i^{th}$ class emitter's $q_i{}^{th}$ periodic sequences with a period of $T_i$. It can be observed that:

1) Both in the train set and the test set, the PRI values of each class exhibit jitter across different cycles.

2) Classes 1, 5, and 8 exhibit staggered pulse repetition frequency (PRF) along with jitter within their cycles, while Classes 2, 4, 6, 7, and 9 show swept PRF. Classes 3 and 10 do not exhibit any discernible patterns in their PRI features.

3) In the train set, the periods of Classes 1∼10 are mostly inconsistent, being 69, 10, non-periodic, 4, 52, 8, 10, 570, 6, and non-periodic respectively. Compared to the train set, the period of Class 1 increases in the test set, while Classes 5 and 8 decrease, yet still showing substantial differences compared to other classes.

4) Although Classes 2 and 7 have the same period, and Classes 3 and 10 lack periodicities, the range of PRI numerical changes differs among these classes, providing potential for recognition.

In summary, the emitter class can be recognized by utilizing the periodicity and amplitude differences of the PRI. Furthermore, even after sorting using the DOA and RF features, there may still be partial class aliasing, and the periodic difference of the PRI provides a possibility for purifying the aliasing samples. It is important to note that while the test set is taken into consideration when selecting PRI as a recog-

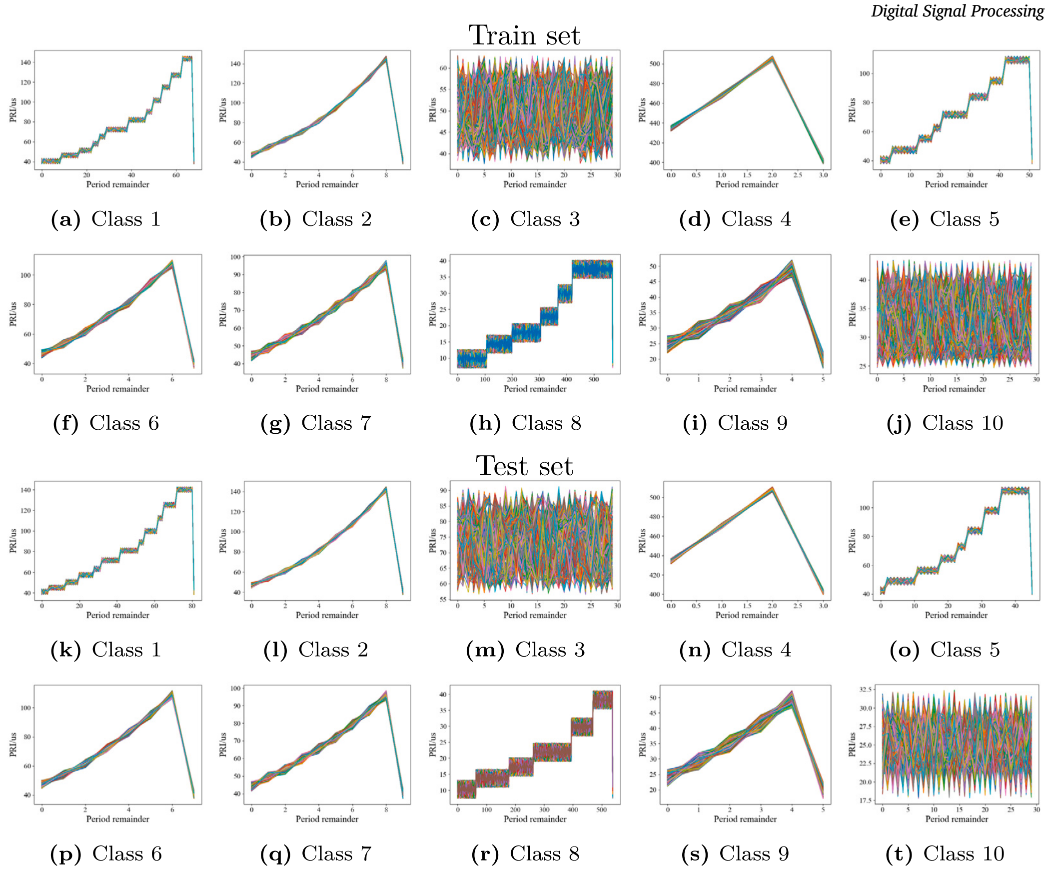



**Fig. 2.** The PRI periodic sequence diagrams for 10 emitter classes.

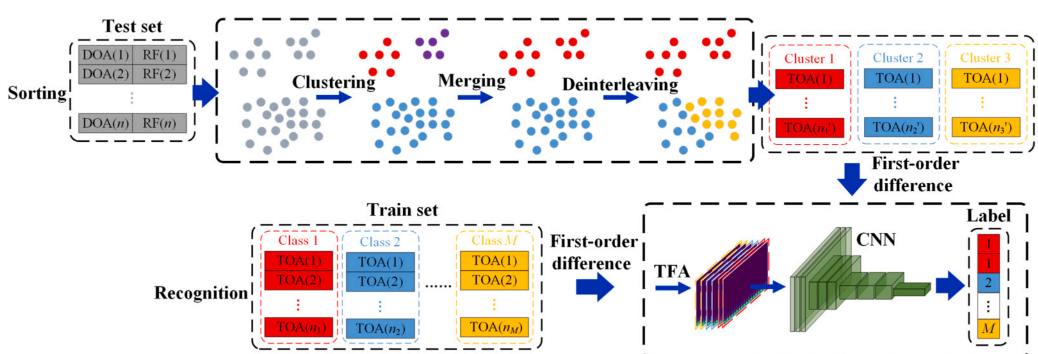



**Fig. 3.** The overall framework of emitter sorting and recognition method based on hierarchical clustering and TFCN.

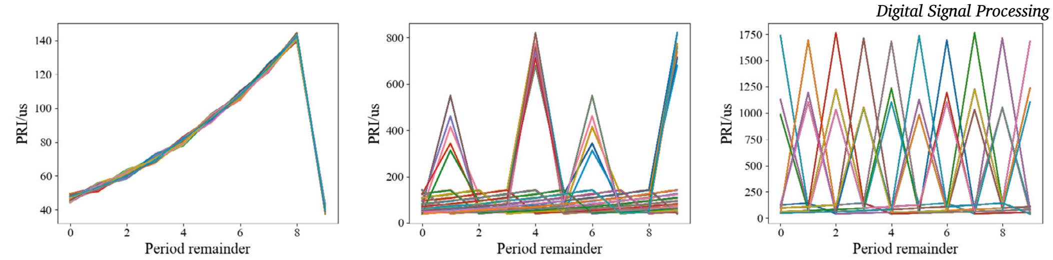
nition feature, this feature is also intrinsically linked to aspects such as inter-radar mutual interference avoidance, combat missions, operational performance metrics. Therefore, it exhibits natural consistency and distinguishability, which are applicable in most scenarios [35–37].

## 3. A multi-radar emitter sorting and recognition method

In the field of sequential data recognition, drawing on the success of Recurrent Neural Networks (RNN) in natural language processing, various LSTM networks have been attempted for emitter recognition with some success [38,39]. While LSTM networks are designed to tackle the challenges of long-sequence prediction, they can still encounter issues with vanishing or exploding gradients when processing data from Class 8 that is characterized by exceptionally long periods. Moreover, when faced with the non-periodic patterns of Classes 3 and 10 as depicted in Fig. 2, the network may struggle to converge.

Considering that time-frequency images can reflect the periodic changes in time series, this paper innovatively applies TFA to PRI features, transforming the 1D time series into 2D time-frequency images, and then using CNNs to complete the emitter recognition. Given that the samples of different classes in the test set are discontinuous in time and overlap, it is almost impossible to extract a fixed-length single-class PRI sample for recognition. Therefore, sorting is required before the recognition process.

Based on the above analysis, this paper proposes an emitter sorting and recognition method based on hierarchical clustering and TFCN. The overall algorithm flow is illustrated in Fig. 3. In the training phase, the first-order differences of the TOA for $M$ class emitters are first calculated to obtain the PRI. Then, the PRI is segmented according to the recognition window length $N_{\mathrm{rec}}$, and TFA is performed on each segment to obtain time-frequency images, which are input into the neural network for training. In the testing phase, for a multi-class mixed test set, all the data are firstly down-sampled uniformly by $N_s$-fold, and KDE is




**(a)** The original cluster **(b)** After clustering, Cluster 2-A **(c)** After clustering, Cluster 2-B

**Fig. 4.** The changes of Class 2 in PRI periodic sequence diagram due to RF clustering.

used to plot the probability density distribution of DOA-RF. Based on maximum a posteriori probability, the mixed samples in the test set are clustered into $C$ clusters. On this basis, clusters belonging to the same class but initially clustered separately are merged according to similarity measured by KLD. Finally, templates are extracted from a small number of pure segments within the aliasing cluster based on the periodicity of PRI, and the aliasing samples are separate through template matching. After sorting, all samples in each cluster are sorted by TOA in ascending order, and PRI is calculated. TFA is then performed according to the recognition window length $N_{\mathrm{rec}}$, and the time-frequency images are input into the trained classifier for recognition.

It is worth noting that the merging should be done before deinterleaving. Intra-class clustering might cause the PRI to lose its periodicity, which could render deinterleaving methods based on PRI periodicity ineffective. As shown in Fig. 4, Class 2 loses its periodicity due to RF clustering. Although aliasing can affect the distribution similarity of intra-class clustering, the distribution similarity between a pure cluster and an aliasing cluster of the same class remains higher than that between clusters of different classes. Thus, the clusters belonging to same class can still be merged by adjusting the threshold. In addition, in some extreme cases, multiple iterations of merging and deinterleaving may be beneficial for improving sorting performance. The number of iterations can be appropriately increased subject to the constraints of time complexity.

### *3.1. Hierarchical clustering based on KDE-KLD-TM*

#### *3.1.1. Initial clustering based on KDE and maximum a posteriori probability*

The proposed method utilizes the differences in the joint probability density of DOA-RF among different emitters to perform sample clustering. To improve clustering speed, the test set samples are first down-sampled uniformly by $N_{\mathrm{s}}$ times to obtain the approximate distribution of the 2D features. Then, the KDE method is used to fit the global joint probability density of DOA-RF [40]:

$$f(\mathrm{DOA},\mathrm{RF}) = \frac{1}{N' h_1 h_2}\sum_{i=1}^{N'} K\left(\frac{\mathrm{DOA}-\mathrm{DOA}_i}{h_1},\frac{\mathrm{RF}-\mathrm{RF}_i}{h_2}\right) \tag{1}$$

where, $N'$ is the total number of samples after down-sampling in the test set, $h_1$ and $h_2$ are the window lengths used to smooth DOA and RF respectively, $\mathrm{DOA}_i$ and $\mathrm{RF}_i$ are the DOA and RF of the $i^{th}$ sample, and $K(\cdot)$ is the kernel function, typically a Gaussian kernel function:

$$K\left(u_1,u_2\right) = \frac{1}{2\pi}\exp\left(-\frac{u_1^2+u_2^2}{2}\right) \tag{2}$$

where, $u_1 = \frac{\mathrm{DOA}-\mathrm{DOA}_i}{h_1}$, $u_2 = \frac{\mathrm{RF}-\mathrm{RF}_i}{h_2}$.

Then, a KDE threshold $\eta_{\mathrm{p}}$ is set, and samples above this threshold are clustered based on their continuity, forming the central cluster. Similarly, the KDE method is used to calculate the 2D joint probability density $f_c$ of each central cluster:

$$f_c(\mathrm{DOA},\mathrm{RF}) = \frac{1}{N_c h_1 h_2}\sum_{i=1}^{N_c} K\left(\frac{\mathrm{DOA}-\mathrm{DOA}_{c,i}}{h_1},\frac{\mathrm{RF}-\mathrm{RF}_{c,i}}{h_2}\right) \tag{3}$$

where, $N_c$ is the number of samples in the $c^{th}$ cluster, and $\mathrm{DOA}_{c,i}$ and $\mathrm{RF}_{c,i}$ are the DOA and RF of the $i^{th}$ sample in the $c^{th}$ cluster.

For the remaining samples in the test set, the posterior probability $P_{c,i}$ of their belonging to $c^{th}$ cluster is calculated:

$$\begin{aligned} P_{c,i}\left(c\,|\mathrm{DOA}_i,\mathrm{RF}_i\right) &= f_c\left(\mathrm{DOA}_i,\mathrm{RF}_i\right) \\ &= \frac{1}{N_c h_1 h_2}\sum_{i=1}^{N_c} K\left(\frac{\mathrm{DOA}_i-\mathrm{DOA}_{c,i}}{h_1},\frac{\mathrm{RF}_i-\mathrm{RF}_{c,i}}{h_2}\right) \end{aligned} \tag{4}$$

The cluster $c$ with the highest posterior probability is selected as the cluster to which $i^{th}$ sample belongs:

$$c = \arg\max_c \left\{P_{1,i}, P_{2,i}, \ldots, P_{C,i}\right\} \tag{5}$$

where, $C$ is the total number of clusters.

#### *3.1.2. Intra-class merging based on KLD*

As shown in Fig. 4, Class 2 exhibit intra-class clustering in their RF features, which disrupts the PRI periodicity of these clusters and affects recognition. To address this issue, we calculate the similarity of each cluster based on KLD after the initial clustering, thereby merging clusters within the same class. Considering that PA has similar distributions across different class clusters and PRI shows significant variations between clusters within the same class, while PW performs better in both aspects, as illustrated in Fig. 1(d) and Fig. 5. Therefore, PW is chosen to replace the RF feature, and the KLD of DOA-PW is calculated.

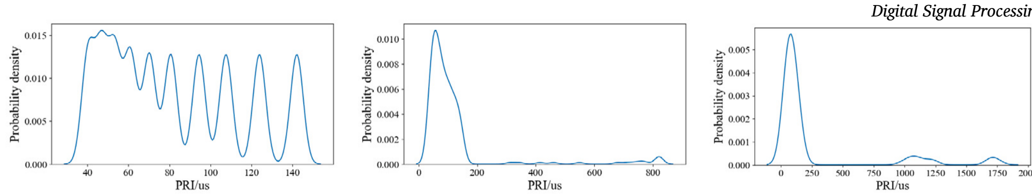
To prevent the erroneous merging of clusters from different classes, before merging, one can determine whether the PRI sample $\left\{\mathbf{PRI}_{c,i}\right\}_{i=1}^{N_c}$ from cluster $c$ exhibits periodicity, which can indicate whether it is the cluster intended for merging. Since the PRI periodic features of most classes are approximately sawtooth wave changes, the periodicity of $\left\{\mathbf{PRI}_{c,i}\right\}_{i=1}^{N_c}$ can be determined by checking if the indexes of the discontinuity points in the sawtooth wave form an arithmetic sequence.

Considering the presence of noise-like PRF jitter of Class 1, 5, and 8, to prevent misjudging jitter points as discontinuity points, a threshold $\eta_{\mathrm{cp},c}$ is set to filter out the jitter points:

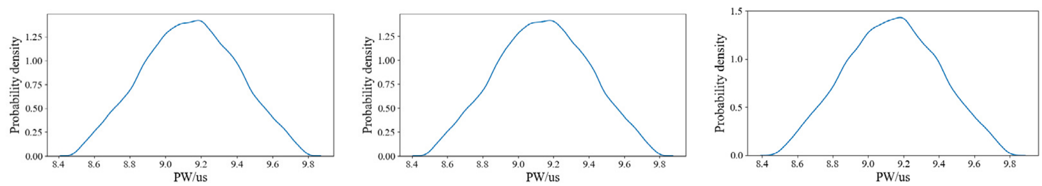
$$\eta_{\mathrm{cp},c} = \frac{\max\limits_{i=1,2,\ldots,N_c}\left|\left\{\Delta^2\mathbf{TOA}_{c,i}\right\}\right|}{2} \tag{6}$$

Select samples that satisfy condition $\left|\left\{\Delta^2\mathbf{TOA}_{c,i}\right\}\right| > \eta_{\mathrm{cp},c}$ to form a discontinuity points sequence $\left\{\mathbf{TOA}_{\mathrm{cp}}^{c,i}\right\}$, and check if the index sequence $\left\{\mathbf{x}_{\mathrm{cp}}^{c,i}\right\}$ of $\left\{\Delta^2\mathbf{TOA}_{\mathrm{cp}}^{c,i}\right\}$ is approximately an arithmetic sequence:

$$\frac{\sum_{i=1}^{N_{\mathrm{cp},c}}\left[\Delta\mathbf{x}_{\mathrm{cp}}^{c,i} = \mathrm{mode}\left(\left\{\Delta\mathbf{x}_{\mathrm{cp}}^{c,i}\right\}\right)\right]}{N_{\mathrm{cp},c}} > \eta_{\mathrm{c}} \tag{7}$$




**(a)** The PRI original distribution **(b)** The PRI distribution of Cluster 2-A after clustering **(c)** The PRI distribution of Cluster 2-B after clustering




**(d)** The PW original distribution **(e)** The PW distribution of Cluster A after clustering **(f)** The PW distribution of Cluster B after clustering

**Fig. 5.** The changes of Class 2 in PRI and PW distribution due to RF clustering.

where, $[\cdot]$ is the indicator function, taking a value of 1 when the internal condition is met, and 0 otherwise. $\mathrm{mode}(\cdot)$ represents taking the mode, $N_{\mathrm{cp},c}$ is the number of discontinuity point samples, and $\eta_{\mathrm{c}}$ is the periodicity determination threshold.

If a cluster has periodicity, it is directly used for recognition. Otherwise, similarity is calculated for merging. The similarity between cluster $p$ and cluster $q$ is defined as:

$$D(p,q)=\frac{D_{KL}(p\,\|\,q)+D_{KL}(q\,\|\,p)}{2} \tag{8}$$

where, $D_{KL}(\cdot)$ denotes the KLD, and since it is not symmetric, the mean of the KLD is taken as the similarity.

The joint KLD of DOA-PW is defined as:

$$D_{KL}(p\,\|\,q)=\int\limits_{\mathrm{DOA}}\int\limits_{\mathrm{PW}} f_p(\mathrm{DOA},\mathrm{PW})\log\left(\frac{f_p(\mathrm{DOA},\mathrm{PW})}{f_q(\mathrm{DOA},\mathrm{PW})}\right)\mathrm{dDOAdPW} \tag{9}$$

If $D(p,q)$ is less than the KLD threshold $\eta_{\mathrm{s}}$, it is considered to be in the same cluster, and a merge is performed.

#### 3.1.3. Inter-class deinterleaving based on template matching

Similar to the issues caused by intra-class clustering, inter-class aliasing can also affect the periodicity of PRI, thereby impacting recognition performance. For cases where all features of multiple classes in a cluster are partially aliasing, a template-matching-based inter-class deinterleaving method is proposed. First, the sample density variation is calculated along the TOA dimension to determine whether there is aliasing in the clusters that have been clustered. If aliasing exists, the pure sample segments within the cluster are further located. Then, estimates of PRI cycle templates are extracted from the pure sample segments, and template matching is performed in the aliasing samples to achieve the effect of deinterleaving.

1) Aliasing judgment

The window length $W_{\rho}$ is set in the TOA dimension, and a unit step sliding window is used to calculate the sample density of the $\omega^{th}$ window of cluster $c$:

$$\rho_{c,\omega}=\frac{W_{\rho}}{\mathrm{TOA}_{c,\omega+W_{\rho}}-\mathrm{TOA}_{c,\omega}},\quad 1\le\omega\le N_c-W_{\rho} \tag{10}$$

A threshold $\eta_{\mathrm{d}}$ is set, and if $\left|\rho_{c,w+1}-\rho_{c,w}\right|\ge\eta_{\mathrm{d}}\min\limits_{w=1,2,\ldots}\left\{\rho_{c,w}\right\}$, aliasing occurs within the window. Further, the aliasing cluster can be divided into aliasing segments $\left\{\mathbf{TOA}_{c,a}\right\}_{a=1}^{N_{\mathrm{a}}}$ and pure segments $\left\{\mathbf{TOA}_{c,p}\right\}_{p=1}^{N_{\mathrm{p}}}$, where $N_{\mathrm{a}}$ and $N_{\mathrm{p}}$ represent the total number of aliasing segments and pure segments, respectively.

2) Template extraction

According to (6) and (7), it is determined whether the pure data segment exhibits periodicity. If so, the period $T_{c,p}$ is estimated:

$$T_{c,p}=\mathrm{mode}\left(\left\{\Delta\mathbf{x}_{\mathrm{cp}}^{c,i}\right\}\right),\quad i=1,2,\ldots,N_{c,p} \tag{11}$$

where, $N_{c,p}$ represents the total number of samples in the $p^{th}$ pure segment of the $c^{th}$ cluster.

The pure data segments are folded according to the period $T_{c,p}$, and the periodic template $\mathrm{PRI}_{\mathrm{model}}^{c,p,i}$ is obtained by calculating the mean:

$$\mathrm{PRI}_{\mathrm{model}}^{c,p,i}=\frac{1}{N_{\mathrm{T}}^{c,p}}\sum_{n=0}^{N_{\mathrm{T}}^{c,p}-1}\mathrm{PRI}_{i+T_{c,p}n},\quad i=1,2,\ldots,T_{c,p} \tag{12}$$

where, $N_{\mathrm{T}}^{c,p}=N_{c,p}/T_{c,p}$ is the number of periods.

3) Template matching

Based on the extracted templates, deinterleaving is performed on the samples in the aliasing segment.

Step 1: The starting index $k$ for matching is determined. First, the aliasing segment is traversed, and for each TOA, the template is matched in the increasing TOA direction. Then the sample corresponding to the minimum loss is identified as the starting index for forward matching.

$$k=\arg\min_{k}\sum_{i=0}^{T_{c,p}-1}\left(\mathrm{PRI}_{\mathrm{model}}^{c,p,i+k}-\mathrm{PRI}_{\mathrm{match}}^{c,a,i+k}\right),\quad k=1,2,\ldots,N_{c,a}+1-T_{c,p} \tag{13}$$

where, $\mathrm{PRI}_{\mathrm{match}}^{c,a,i}=\min\left|\mathrm{PRI}_{c,a,j}-\mathrm{PRI}_{\mathrm{model}}^{c,p,i}\right|,\quad j=1,2,\ldots,N_{c,a}$. $N_{c,a}$ represents the total number of samples in the $a^{th}$ aliasing segment of cluster $c^{th}$.

Step 2: Forward matching starts from $\mathrm{TOA}_{c,a,k}$. Given that the starting index for forward matching of a certain class may not be at the first period segment of that class within the aliasing segment, there might be parts of the aliasing segment for that class that are not matched. Therefore, after completing the forward matching, the aliasing segment and the template are flipped, and backward matching starts from the starting index of the forward match to ensure that the entire class is separated from the aliasing segment.

Step 3: Steps 1 and 2 are repeated until all classes in the current aliasing segment have been matched with their respective periodic templates.

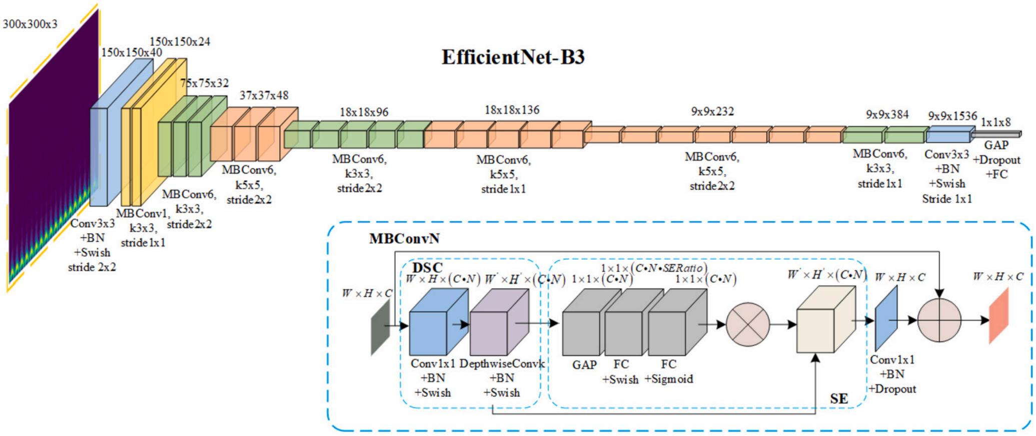



**Fig. 6.** The EfficientNet-B3 network architecture diagram.

### *3.2. Emitter recognition based on TFCN*

After completing the sorting of the aliased emitters, the purified samples are sorted according to the TOA from smallest to largest, and the PRI is calculated. The TFA is then performed on the PRI values, and the time-frequency images are input into CNN for recognition. During the training phase, data augmentation is performed on the input data to enhance the robustness of the algorithm and reduce the impact of sorting errors on recognition.

#### *3.2.1. Data augmentation*

Each class in the train set is divided into segments with a recognition window length $N_{\text{rec}}$. For each segment of the TOA or PRI sequences, the following data augmentation procedures are applied.

1) Noise Adding: A normally distributed noise sequence with a mean of 0 and a standard deviation of $\sigma_n$ is added to the PRI sequence. Clipping is applied to ensure that the enhanced PRI sequence remains a positive number sequence.

2) Noise Mixing: A random selection of consecutive samples from the PRI sequence is chosen, and the values of these selected samples are replaced with noise. The noise is generated from a normal distribution with the same mean and variance as the original PRI sequence. Clipping is applied to ensure that the enhanced PRI sequence remains a positive number sequence.

3) End-Zeroing: A random selection of consecutive samples at the end of the PRI sequence is chosen, and the values of these selected samples are replaced with 0.

4) Reversing: The PRI sequence is reversed in order.

5) Interpolating and Loss: A random selection of samples is inserted into the TOA sequence, and an equal number of samples is randomly removed. The values of the samples to be inserted are set between the values of adjacent samples.

6) Multi-Class Mixing: A TOA sequence of a certain class is mixed with the TOA sequences of the other classes in a specified quantity. Samples exceeding the length $N_{\text{rec}}$ are truncated.

#### *3.2.2. Time-frequency analysis*

Fig. 2 illustrates that for the 8 classes where the PRI exhibits periodicity, the changes in PRI are approximately characterized by a sawtooth wave pattern, meaning they satisfy:

$$f_i(t) = \frac{A_i}{T_i} t, \quad 0 \le t \le T_i, \quad i = 1, 2, \dots, 8 \tag{14}$$

where, $A_i$ and $T_i$ represent the amplitude and period of the PRI for the $i^{th}$ class, respectively.

The Fourier series expansion of (14) is:

$$F_i(t) = \frac{A_i}{2} + \sum\nolimits_{n=1}^{\infty} \frac{A_i}{\pi n} \sin\left(\frac{2\pi n}{T_i} t\right) \tag{15}$$

It can be seen that the spectrum is composed of a DC component, a sinusoidal fundamental wave component, and higher-order harmonic components with decreasing amplitudes. Different PRI periods correspond to different fundamental and harmonic frequencies, thus providing a new dimension for recognition.

Based on the above analysis, considering that time-frequency transformations can simultaneously represent time and frequency characteristics and hold potential benefits for image recognition tasks, time-frequency images are selected as the recognition features. Since sorted samples may still exhibit minor multi-class aliasing, to avoid cross-term interference, the computationally efficient STFT is chosen for TFA. The definition of the STFT is as follows:

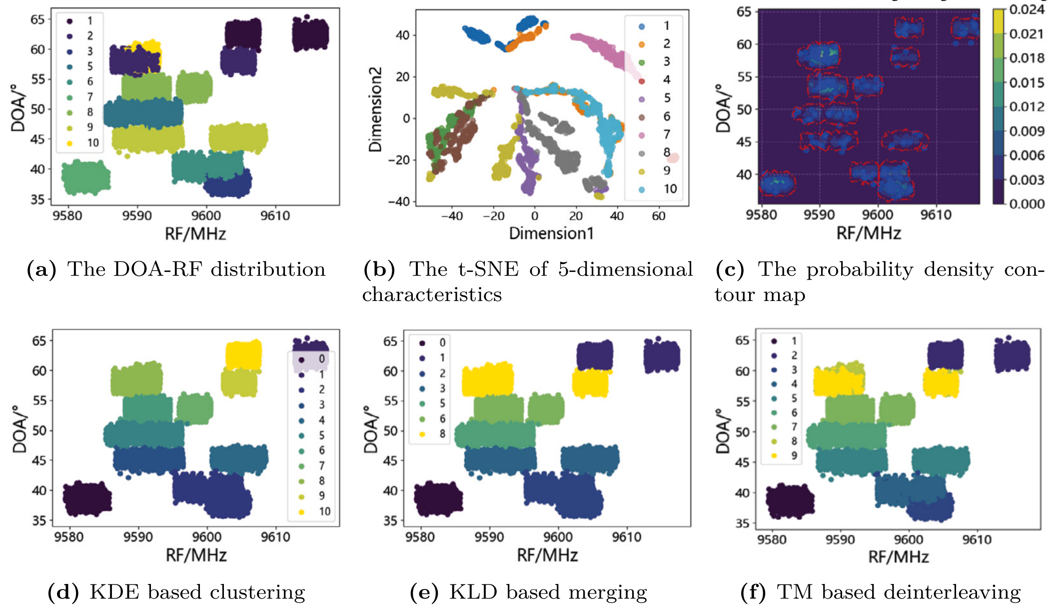
$$\text{STFT}_x(T, F) = \sum\nolimits_{n=T-W_{\text{stft}}/2}^{T+W_{\text{stft}}/2} x(n)\,\omega(n-T)\,e^{-j\frac{2\pi}{N_{\text{fft}}} Fn} \tag{16}$$

where, $x(n)$ represents the discrete samples to be processed, $\omega(n)$ is the window function, $W_{\text{stft}}$ is the window length of STFT, $N_{\text{fft}}$ is the number of points for the Fourier transform, and $T$ and $F$ represent the time and frequency sampling units, respectively. The window function used in this paper is the Hanning window, and the window length is adjusted based on the periodicity of the PRI.

#### *3.2.3. CNN framework*

EfficientNet is selected as the CNN framework for classification due to its state-of-the-art (SOTA) classification performance on the ImageNet dataset in 2019 [41]. To optimize recognition performance under limited resources, the model scales the depth, width, and input image resolution according to a compound scaling strategy, yielding eight versions of the model (EfficientNet-B0 to EfficientNet-B7) with progressively increasing sizes. Considering both computational speed and accuracy, EfficientNet-B3 is selected for this study, and its network structure is illustrated in Fig. 6.

The network utilizes stacked Mobile Inverted Bottleneck Convolutions (MBConv) at varying scales for multi-scale feature extraction and fusion. Each MBConv block comprises two main components: Depthwise Separable Convolution (DSC) and Squeeze-and-Excitation (SE). The DSC module first performs feature fusion across channels and then applies convolution to each channel individually, thereby significantly improving computational efficiency compared to standard convolution. The SE module adjusts the weights of each channel, enhancing the extraction of channel correlation information and implementing a channel attention mechanism. Finally, spatial features are aggregated through global average pooling, and a fully connected layer adjusts the output dimensions for recognition.




(a) The DOA-RF distribution (b) The t-SNE of 5-dimensional characteristics (c) The probability density contour map

(d) KDE based clustering (e) KLD based merging (f) TM based deinterleaving

**Fig. 7.** The characteristic distributions, density clustering map and sorting results of the test set.

**Table 1**
The algorithm parameter settings.

| Parameters | Values |
|---|---|
| Down-sampling factor $N_s$ | 50 |
| KDE threshold $\eta_p$ | 0.001 |
| Periodicity discrimination threshold $\eta_c$ | 0.3 |
| KLD threshold $\eta_s$ | 2 |
| Aliasing discrimination threshold $\eta_d$ | 1.5 |
| Recognition window length $N_{rec}$ | 1024 |
| STFT window length $W_{stft}$ | 64 |
| The number of points for the Fourier transform | 1024 |
| Learning rate $l_r$ | $10^{-5}$ |
| Batch size | 4 |
| Epoch | 20 |

## 4. Simulations and results

### 4.1. Parameter setting and evaluation metric

The algorithm parameter settings are shown in Table 1. The down-sampling factor strikes a balance between the algorithm's processing speed and clustering effectiveness. The KDE threshold and KLD threshold modulate the strength of clustering and merging. The periodicity discrimination threshold and aliasing discrimination threshold are used to select clusters that require merging or deinterleaving. Parameters related to the Fourier transform directly affect the PRI time-frequency features. These parameter settings influence the sorting and recognition performance and should be dynamically adjusted according to practical scenarios. In this paper, a relatively optimal combination of parameters is selected through multiple simulations.

The sorting accuracy is evaluated using the Rand Index RI, which measures the overlap between two class clusters [42]:

$$RI = \frac{TP + TN}{TP + TN + FP + FN} \tag{17}$$

where, TP, TN, FP and FN respectively represent the number of cases where samples of the same class are in the same cluster, samples of different classes are in separate clusters, samples of different classes are in the same cluster, and samples of the same class are in separate clusters.

Without loss of generality, the recognition performance is evaluated using the confusion matrix and accuracy [43].

### 4.2. Scenario without pulse loss

#### 4.2.1. Analysis of sorting result

Fig. 7 displays the DOA-RF distribution, the t-Distributed Stochastic Neighbor Embedding (t-SNE) visualization of the 5-dimensional features, the probability density contour map, and the results of clustering, merging, and deinterleaving. Since Class 4 has a significantly different RF distribution range compared to other classes, making it easily separable, this class is omitted from the distribution and clustering results for visual clarity. As depicted in Fig. 7(a), the 2D features of all classes exhibit horizontal strip distributions. Except for Class 7, the feature distributions of the classes show some degree of overlap, with particularly large overlaps between Classes 3 and 6, as well as between Classes 2 and 10. Additionally, Classes 1, 2, 8, and 9 exhibit feature clustering. It should be noted that even with all 5-dimensional features, the aforementioned classes still exhibit aliasing and clustering, as shown in Fig. 7(b).

Fig. 7(c) intuitively presents the probability density changes of the 2-dimensional features of the test set samples. In this figure, the background color represents the density, with blue dots indicating the down-sampled samples and red dashed lines marking the KDE threshold. Clusters are divided according to the KDE threshold on the probability density contour map, and the posterior probabilities of the samples belonging to each cluster are calculated. The samples are assigned to the cluster with the highest posterior probability. The clustering result is shown in Fig. 7(d), with a total of 11 clusters. Classes 1, 2, 8, and 9, which exhibit feature clustering, are incorrectly divided into two clusters each. Classes 2 and 10, as well as classes 3 and 6, which display feature aliasing, are erroneously merged into a single cluster. The Rand Index is 70.89%.

Following the merging process, the 11 initial clusters are consolidated into 7, with the Rand Index increasing to 82%, achieving the merging of intra-class clustering for 4 classes, as shown in Fig. 7(e). After deinterleaving, the 7 clusters are further refined into 9, and the Rand Index improves to 96.22%, effectively resolving feature aliasing for 4 classes, as shown in Fig. 7(f). The main source of sorting error stems from inaccurate division of edge aliased samples.

The comparative method selected is HACOT [9]. This method is a bottom-up hierarchical clustering approach that starts by creating as many clusters as possible and then iteratively merges them based on inter-cluster similarity, until the similarity between clusters falls below a threshold. Consequently, this algorithm is time-consuming and

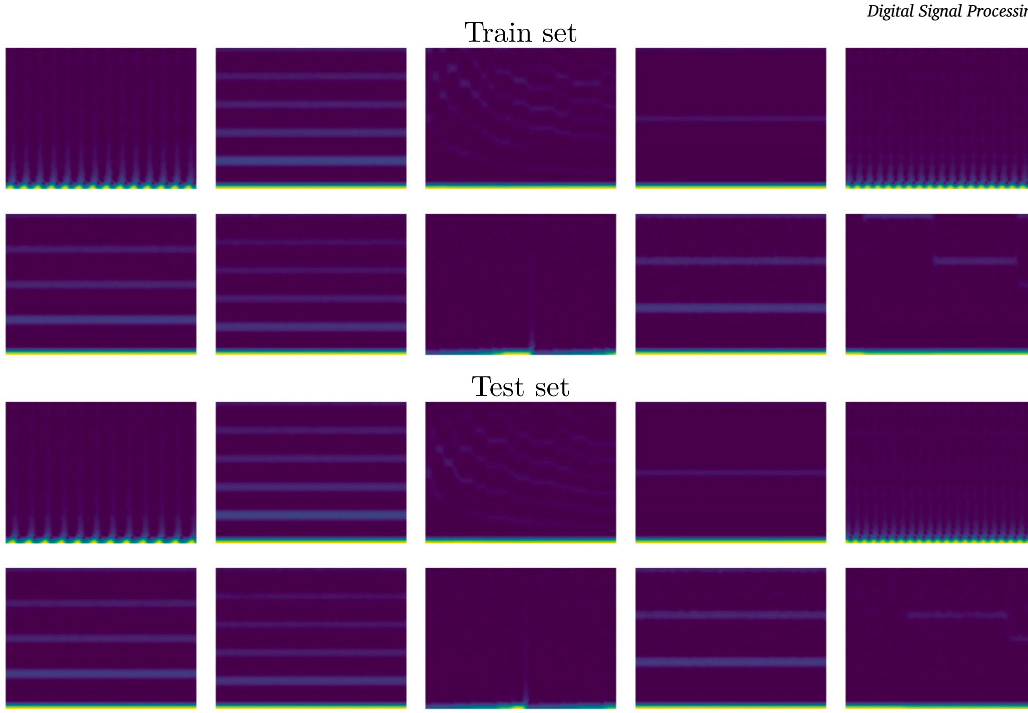



**Fig. 8.** The time-frequency image examples for train and test set.

**Table 2**
The comparison of sorting performance.

| Methods | Computation time | Rand Index |
|---|---|---|
| HACOT | 224.3 s | 83.65% |
| KDE | **5.97 s** | 70.89% |
| KDE-KLD | 7.72 s | 82% |
| KDE-KLD-TM | 58.61 s | **96.22%** |

has difficulty handling inter-class aliasing issues. The comparison results are shown in Table 2. KDE-KLD achieves both inter-class clustering and intra-class merging, matching the sorting performance of the HACOT method while significantly improving processing speed by approximately 29 times, reducing the runtime to a matter of seconds. With the addition of deinterleaving, the processing speed is improved by about 3.8 times compared to the HACOT method, taking only 58.61 seconds to sort 2.06 million samples, and the Rand Index is increased by 12.57%. Therefore, the proposed hierarchical clustering algorithm based on KDE-KLD-TM can effectively address feature aliasing issues and offers faster computation time.

#### 4.2.2. *Analysis of recognition result*

The time-frequency images for the 10 emitter classes corresponding to Fig. 2 are shown in Fig. 8, with more obvious differences between the classes.

1) For Classes 2, 4, 6, 7, and 9, where the window length $W_{\text{stft}}$ can cover at least two complete periodicities, the time-frequency image features are consistent with the analysis of the sawtooth wave characteristics presented previously. The features remain constant over time and consist of a DC component, a fundamental wave component, and higher-order harmonic components with decreasing amplitudes. The primary differences among these classes lie in the frequencies or amplitudes of the harmonics.

2) For Classes 1, 5, and 8, where the window length $W_{\text{stft}}$ cannot cover two complete periodicities, the time-frequency images consistently exhibit a DC component whose amplitude varies gradually over time. The key distinctions among these classes lie in the different periods of amplitude variation.

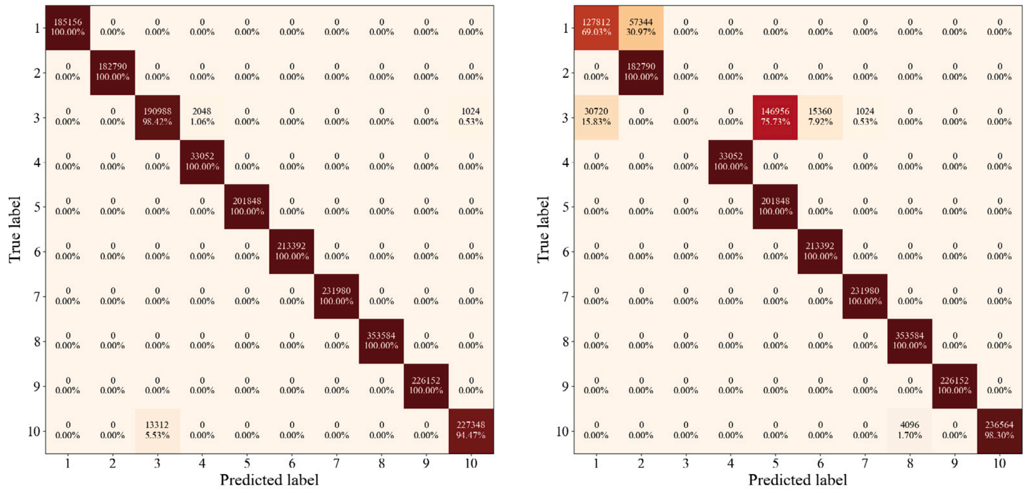
3) For Classes 3 and 10, which lack overall periodic features, TFA reveals local periodicity and a strong DC component. Class 3 shows varying combinations of frequency components across different time intervals, each with distinct durations. Class 10 exhibits different single frequencies at various time intervals, revealing characteristics of sinusoidal swept PRF.

First, we analyze the recognition performance without considering mis-sorting. For comparison, we select a temporal convolution method, DP-ATCN [34], which processes PRI sequences instead of time-frequency images. To ensure a fair comparison, we maintained the same input data length and optimized the network parameters to their best possible settings. The confusion matrix comparison is shown in Fig. 9. The x-axis represents the predicted class, and the y-axis represents the true class. The diagonal values represent recall rates, and the sum of percentages in each row of the matrix equals 1. The recognition accuracies of the TFCN and DP-ATCN are 99.21% and 87.61%, respectively, with the proposed method improving accuracy by 11.60%.

The proposed method achieves recall rates of 98.42% and 94.47% for Classes 3 and 10, respectively, which lack periodicity. For the other classes, the recall rates reach 100%. These results demonstrate the superior recognition performance of the proposed method for periodic sequences, as well as its potential for recognizing sequences without periodicity. It should be noted that to improve anti-jamming performance and reduce ambiguity, common radar PRF variation strategies, such as staggered frequency and swept frequency, possess certain periodic characteristics.

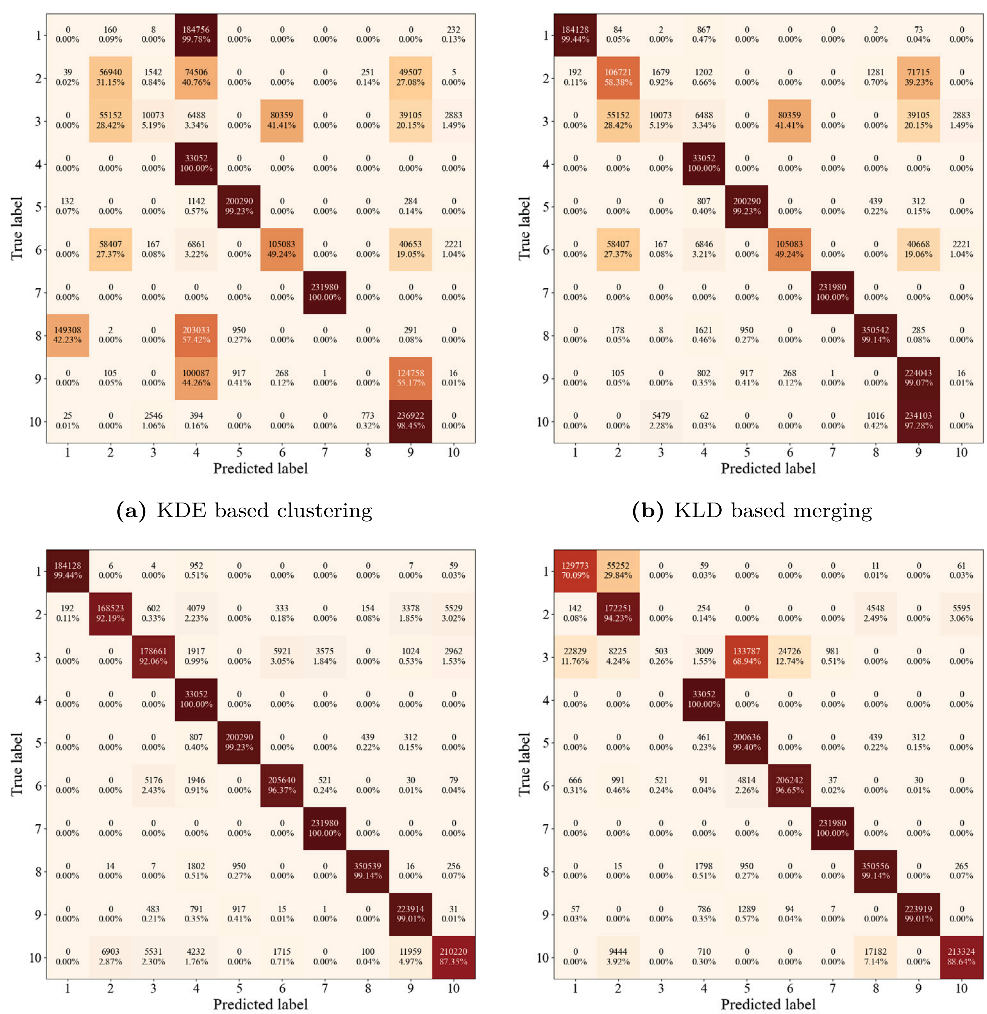
The DP-ATCN method considers both feature distribution range and periodicity. Therefore, the significant difference in the distribution range of Class 3 between the training and testing sets results in the method being unable to correctly identify Class 3 at all. Additionally, although Classes 1 and 2 have differences in periodicity, their similar distribution ranges cause the DP-ATCN to misidentify approximately 1/3 of Class 1 as Class 2. These findings suggest that the feature distribution range may not be a highly effective or reliable feature for emitter recognition, and classifiers should not overly rely on this feature.

In conjunction with the proposed sorting method, the recognition results are illustrated in Fig. 10. When only KDE clustering is performed,




(a) TFCN (b) DP-ATCN

**Fig. 9.** The recognition results under supervised sorting conditions.




(a) KDE based clustering (b) KLD based merging

(c) TM based deinterleaving (d) DP-ATCN

**Fig. 10.** The impact of sorting steps on recognition performance.

the confusion matrix is depicted in Fig. 10(a), with a recognition accuracy of 36.95%. The focus is on the impact of inter-class aliasing and intra-class clustering on recognition performance.

1) For classes with severe aliasing, including Classes 3 and 6 and Classes 2 and 10, the recall rates are all less than 50%, specifically 5.19%, 49.24%, 31.15%, and 0%, respectively.

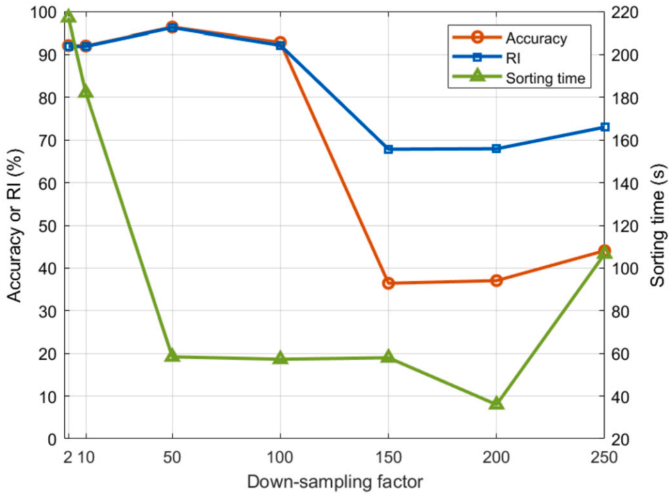



**Fig. 11.** The accuracy and processing speed under different down-sampling multiples.

2) For classes exhibiting feature clustering, including Classes 1, 2, 8, and 9, the recall rates are all less than 60%, specifically 0%, 31.15%, 0%, and 55.17%, respectively.

It can be seen that both feature aliasing and feature clustering can severely degrade recognition performance.

With the addition of merging processing, the recognition accuracy increases by 33.15%, reaching 70.10%, as shown in Fig. 10(b). For classes exhibiting feature clustering, including Classes 1, 2, 8, and 9, the recall rates increase to 99.44%, 58.38%, 99.14%, and 99.07%, respectively, improving by 9.14% to 99.07%. This demonstrates that merging processing can effectively improve recognition performance. However, for Class 2, due to the presence of aliasing, the improvement in recall rate is limited.

With the further addition of deinterleaving processing, the recognition accuracy increases by another 26.23%, reaching 96.33%, as shown in Fig. 10(c). For classes with aliasing, including Classes 3 and 6 and Classes 2 and 10, the recall rates increase to 92.06%, 96.37%, 92.19%, and 87.35%, respectively, improving by 47.13% to 87.35%. This demonstrates that the deinterleaving algorithm can also effectively improve recognition performance. For Class 10, however, the recall rate remains below 90% due to unclear periodic features and inaccuracies in sorting edge-aliased samples, which affect the time-frequency features.

Fig. 10(d) shows the confusion matrix for the proposed sorting method based on KDE-KLD-TM combined with DP-ATCN, with a recognition accuracy of 85.43%, which is a decrease of 2.18% and 10.90% compared to the supervised sorting and TFCN, respectively.

Combining the KDE, KDE-KLD, KDE-KLD-TM and HACOT sorting methods with the TFCN and DP-ATCN recognition methods, the comparison of recognition performance is presented in Table 3. It can be seen that both aliasing and clustering reduce the accuracy of both recognition methods. The proposed method exhibits the best performance among the sorting and recognition combinations, with an accuracy improvement of 27.78% compared to the baseline method. Additionally, two notable comparisons are observed: HACOT+TFCN versus KDE-KLD+TFCN, and HACOT+DP-ATCN versus HACOT+TFCN.

The recognition accuracy of HACOT+TFCN is 14.2% lower than that of KDE-KLD+TFCN, despite a difference in sorting performance of only 1.65%. This discrepancy can be attributed to the fact that although HACOT possesses merging capabilities, it does not perform intra-class merging for Class 2, which exhibits both aliasing and clustering. Compared to KDE-KLD, the TP in the numerator of the Rand Index decreases, and the TN increases, so the total sum remains almost unchanged. However, the lack of intra-class merging causes a significant decrease in the recognition performance of Class 2, which indirectly affects the

**Table 3**
The recognition performance under different combinations of sorting and recognition methods.

| Methods | Recognition accuracies |
|---|---|
| HACOT+DP-ATCN | 68.55% |
| HACOT+TFCN | 55.90% |
| KDE+TFCN | 36.95% |
| KDE-KLD+TFCN | 70.10% |
| KDE-KLD-TM+DP-ATCN | 85.43% |
| KDE-KLD-TM+TFCN | **96.33%** |

recognition performance of other classes, thereby reducing the overall recognition accuracy. The recognition accuracy of HACOT+DP-ATCN is 12.65% higher than that of HACOT+TFCN, for reasons consistent with the previous analysis. DP-ATCN is less sensitive to periodicity than TFCN, allowing it to retain some recognition capability for classes with aliasing.

The down-sampling factor used during the construction of the center cluster is a primary parameter affecting both accuracy and processing speed. Fig. 11 illustrates the impact of different down-sampling factors on accuracy, Rand Index, and sorting time. It can be seen that when the down-sampling factor is between 2 and 100, the recognition accuracy remains above 90%, and the processing time reaches a relatively low value of around 58 seconds at a down-sampling factor between 50 and 100. As the down-sampling factor continues to increase, the recognition accuracy rapidly deteriorates to around 40%. This is because excessive down-sampling results in the loss of most sample points, making it impossible to form effective central clusters. Consequently, the assignment of all samples becomes highly unreliable, leading to more complex intra-class clustering and inter-class aliasing. This further complicates subsequent merging and deinterleaving processes, ultimately worsening the recognition performance and potentially increasing processing time due to need for frequent deinterleaving. Therefore, the down-sampling factor should not be too large in order to maintain good recognition performance while reducing processing time.

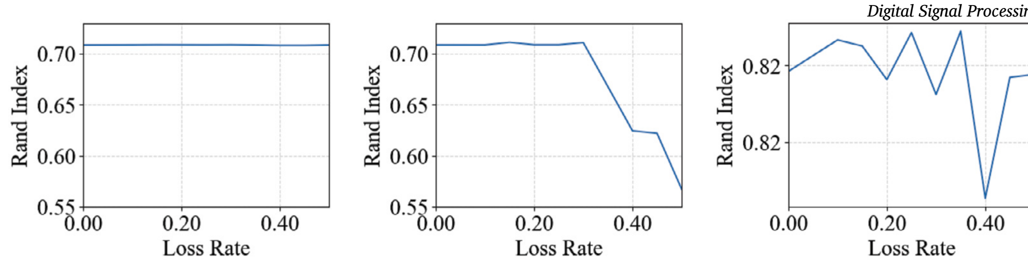
### *4.3. Scenario with pulse loss*

Furthermore, given that random pulse loss is a prevalent issue in real-world applications, the subsequent simulation explores performance variations in sorting, recognition, and their combined. Specifically, this analysis involves gradually increasing the rate of random sample loss within the test sets by 5% intervals, up to a maximum loss rate of 50%.

#### *4.3.1. Analysis of sorting result*

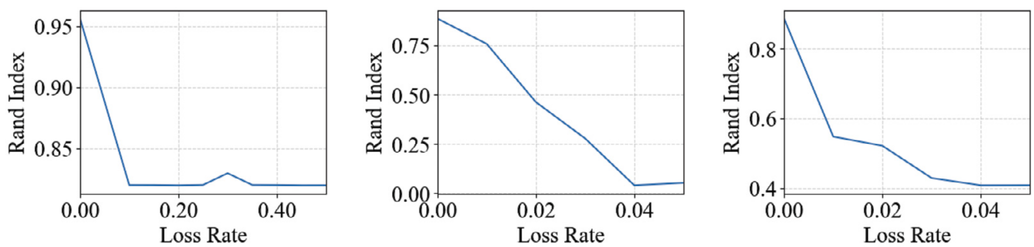
The sorting algorithm introduced in this paper comprises three submodules: clustering, merging and deinterleaving. In the subsequent analysis, we systematically examine the impact of random pulse loss on each of these components, evaluating their performance and robustness under conditions of pulse loss.

The clustering module of the proposed algorithm performs down-sampling on the samples prior to clustering. Given that random pulse loss can be viewed as a form of down-sampling, this theoretically endows the module with robustness against such occurrences. As illustrated in Fig. 12(a), the RI stabilizes at approximately 0.7 under a down-sampling factor of 50. When the down-sampling factor is further increased to 100, the variation of RI with respect to the loss rate is depicted in Fig. 12(b). At loss rates between 0% and 30%, the RI remains largely unchanged despite increases in the loss rate; however, when the loss rate exceeds 30% and approaches 50%, the RI begins to decline. This phenomenon can be attributed to the combined effects of higher loss rates and down-sampling factor, which together disrupt the DOA-RF distribution utilized for clustering. Consequently, the proposed algorithm




**(a)** Clustering performance with 50 times down-sampling **(b)** Clustering performance with 100 times down-sampling **(c)** Merging performance

**Fig. 12.** Clustering and merging performance with different loss rates.




**(a)** Random loss of all pulses **(b)** Random loss of pure segments **(c)** Random loss of aliasing segments

**Fig. 13.** Deinterleaving performance at different loss rates.

is designed to adaptively adjust the down-sampling factor. This flexibility allows it to balance the clustering performance degradation caused by pulse loss with the computational efficiency of clustering processing time, thereby optimizing overall performance.

For the merging module, random pulse loss can be regarded as random down-sampling of the DOA-PW distribution. In general, the DOA distribution is close to normal and the PW distribution is close to normal or mixed normal. When the sample size is sufficiently large, random pulse loss exerts little impact on the characteristics of both the PW and DOA distributions. Consequently, the module retains its ability to leverage the similarity in the DOA-PW distribution for merging cluster classes effectively. Fig. 12 demonstrates the variation of the merging performance with the loss rate, and the RI remains virtually constant at loss rate below 50%, which indicates the robustness of the proposed merging algorithm against random pulse loss.

The operation of the deinterleaving module hinges on the matching ability of extracting PRI period template. Pulse loss can severely compromise this periodicity, thereby undermining algorithm performance. As depicted in Fig. 13(a), the RI drops markedly and stabilizes as the loss rate increases, indicating that at a 10% loss rate, the proposed algorithm fails to perform deinterleaving effectively. To provide a more concrete illustration of the impact on performance, we examine two scenarios: pulse loss in pure segments and in aliased segments, using classes 2 and 10 as examples where aliasing occurs. This case allows us to assess how sensitive the template extraction and matching submodules within the deinterleaving process are to pulse loss. Fig. 13(b) reveals that when pulses are lost from pure segments, the RI declines sharply with increasing loss rates, nearly reaching zero by a 4% loss rate. This decline signifies that the extracted template no longer correctly matches the aliased segments. Similarly, Fig. 13(c) shows that when pulses are lost from aliased segments, the RI exhibits a comparable trend in response to varying loss rates. In conclusion, these findings underscore the high sensitivity of the proposed deinterleaving module to pulse loss, highlighting its critical dependence on accurate template extraction and matching for optimal performance.

#### *4.3.2. Analysis of recognition result*

Fig. 14 illustrates the periodic sequence and time-frequency representation of the test set following a 10% pulse loss within a single period, and for the non-periodic Classes 3 and 10 only 20 points are shown. From the periodic sequence diagram, it is evident that pulse loss can result in incomplete PRI sequences, thereby disrupting the original periodic characteristics or even losing periodicity. Fortunately, when the loss rate remains relatively low, the overall amplitude trend of the sequence can largely preserve its pre-loss characteristics. The impact of pulse loss is also evident in the time-frequency diagram, where the horizontal stripes indicative of periodicity become blurred and discontinuous, which significantly increases the difficulty of recognition methods based on time-frequency features.

Fig. 15 shows the variation in recognition performance with respect to the loss rate for three methods: the ideal sorting + TFCN without data augmentation (termed Ideal+TFCN), the same method but with data augmentation (termed Ideal+ATFCN), and KDE-KLD-TM + TFCN with data augmentation (termed Real+ATFCN). The sorting performance at the corresponding loss rate is also attached. It is evident that overall recognition accuracy declines as the loss rate increases, irrespective of the correctness of the sorting outcome. In scenarios where loss occurs, sorting errors contribute to a deterioration in recognition performance ranging from 8% to 23%. While employing random loss during data augmentation can significantly enhance recognition performance, there remains a performance gap exceeding 10% when compared to conditions without any loss. On a more positive note, for loss rates between 10% and 50%, the reduction in recognition accuracy is less pronounced relative to the increase in pulse loss rate. Even in scenarios with a 50% loss rate, the recognition accuracy remains above 50%.

## 5. Conclusions

To addresses the issues of slow sorting processing speeds, intra-class feature clustering, inter-class feature aliasing, and inadequate attention to effective recognition features in emitter sorting and recognition, this paper proposes a multi-radar emitter sorting and recognition method based on hierarchical clustering and TFCN. First, we conduct a feature

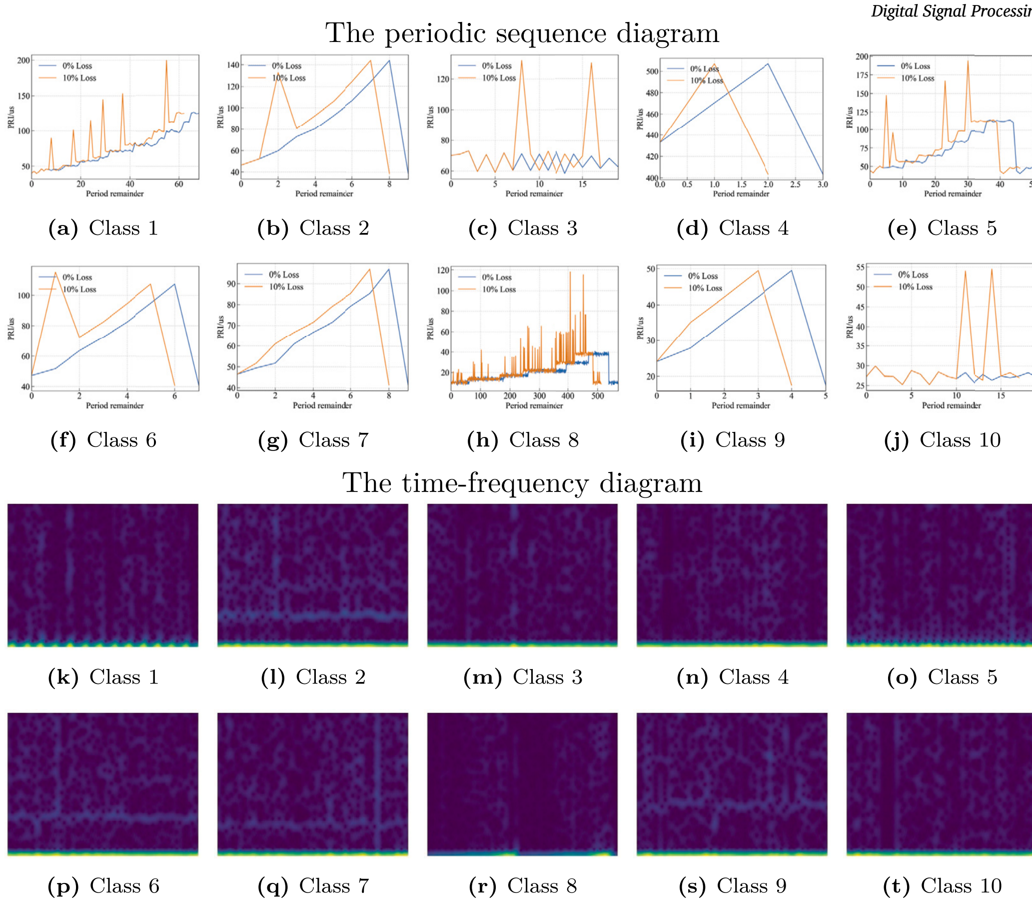



**Fig. 14.** The PRI periodicity and time-frequency patterns of 10 emitter classes for a 10% loss rate.

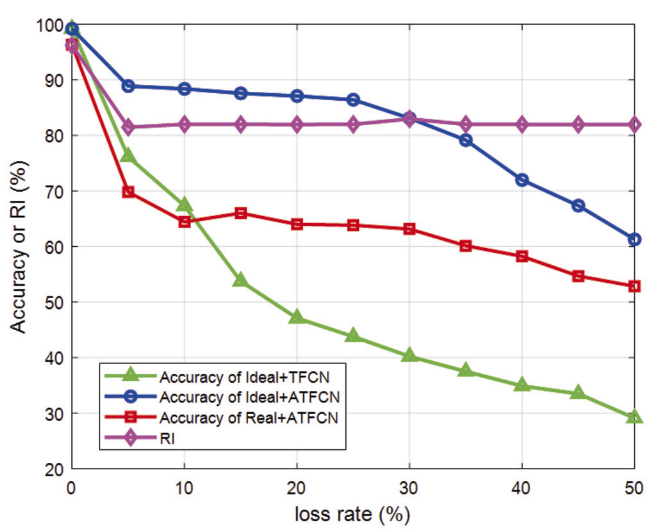



**Fig. 15.** Performance variation with loss rate in pulse loss scenario.

analysis on the external dataset from two perspectives: separability and consistency. Based on this analysis, DOA and RF are selected as sorting features, and PRI is selected as a recognition feature. On this foundation, the probability density function of DOA-RF is estimated using KDE to draw a probability density contour map, and inter-class clustering is performed based on a density threshold. Then, KLD is used to evaluate inter-cluster similarity for intra-class merging. Finally, cycle template matching is applied to purify aliased emitters. For the purified samples, time-frequency features are extracted, and a CNN classifier is used to achieve emitter recognition.

Through testing on a challenging dataset with clustering and aliasing issues, the proposed algorithm can sort 2.06 million samples within 58.61 seconds, achieving a Rand Index of 96.22% and a recognition accuracy of 96.33%, which demonstrates the effectiveness of the selected features and sorting and recognition methods. Compared to the sorting method of HACOT and the recognition method of DP-ATCN, the sorting speed is improved by approximately 3.8 times, with individual metric improvements of 12.57% and 11.6%, respectively. When combined, the recognition accuracy is improved by 27.78%, confirming the superiority of the proposed method.

While our proposed method achieves good recognition results, there is still room for improvement in accurately sorting edge-overlapping samples. Moreover, there are two key limitations to our study:

1) Approximate Monotonicity Assumption for PRI Changes: When merging judgments, it is required that the PRI changes within the periodicity approximately satisfy monotonicity, otherwise false merges may occur.

2) Requirement for Pure Data Segments: If no pure data segment with a complete PRI periodicity can be found within an aliasing cluster, the proposed deinterleaving method will not be effective.

Although such situations are rare in scenarios without pulse loss, the performance of the proposed method does degrade to some extent in complex practical scenarios involving pulse loss. This degradation is primarily due to the disruption of PRI periodicity. Future research could focus on enhancing the algorithm's robustness to address these challenges.

## CRediT authorship contribution statement

**Jiaxiang Zhang:** Conceptualization, Methodology, Writing – original draft. **Bo Wang:** Methodology, Software, Validation, Visualization.

**Xinrui Han:** Methodology, Software, Validation, Visualization. **Min Zhao:** Methodology, Software, Visualization. **Zhennan Liang:** Funding acquisition, Resources, Supervision. **Xinliang Chen:** Formal analysis, Funding acquisition. **Quanhua Liu:** Conceptualization, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

## References

[1] S. Scholl, S. Wagner, End-to-end recognition of interleaved radar emitters from the spectrogram, in: 2023 20th European Radar Conference (EuRAD), IEEE, 2023, pp. 286–289.
[2] S. Wei, Q. Qu, X. Zeng, J. Liang, J. Shi, X. Zhang, Self-attention bi-lstm networks for radar signal modulation recognition, IEEE Trans. Microw. Theory Tech. 69 (2021) 5160–5172.
[3] Z. Zhang, M. Zhu, Y. Li, Y. Li, S. Wang, Joint recognition and parameter estimation of cognitive radar work modes with lstm-transformer, Digit. Signal Process. 140 (2023) 104081.
[4] H. Li, W. Jing, Y. Bai, Radar emitter recognition based on deep learning architecture, in: 2016 CIE International Conference on Radar (RADAR), IEEE, 2016, pp. 1–5.
[5] X. Han, S. Chen, M. Chen, J. Yang, Radar specific emitter identification based on open-selective kernel residual network, Digit. Signal Process. 134 (2023) 103913.
[6] Z. Xiao, Z. Yan, Radar emitter identification based on novel time-frequency spectrum and convolutional neural network, IEEE Commun. Lett. 25 (2021) 2634–2638.
[7] H. Cong Feng, K. Li Jiang, X. Yu Zhao, A. Al-Malahi, B. Tang, Self-supervised contrastive learning for extracting radar word in the hierarchical model of multifunction radar, IEEE Trans. Aerosp. Electron. Syst. 59 (2023) 9621–9634.
[8] B. Park, J.M. Ahn, Intra-pulse modulation recognition using pulse description words and complex waveforms, in: 2017 International Conference on Information and Communication Technology Convergence (ICTC), IEEE, 2017, pp. 555–560.
[9] M. Mottier, G. Chardon, F. Pascal, Deinterleaving radar emitters with optimal transport distances, IEEE Trans. Aerosp. Electron. Syst. 60 (2024) 3639–3651.
[10] S. Scholl, S. Brüggenwirth, Incremental deinterleaving of radar emitters, IEEE Aerosp. Electron. Syst. Mag. 38 (2023) 26–36.
[11] X. Feng, X. Hu, Y. Liu, Radar signal sorting algorithm of k-means clustering based on data field, in: 2017 3rd IEEE International Conference on Computer and Communications (ICCC), IEEE, 2017, pp. 2262–2266.
[12] L. Lesieur, J.-M. Le Caillec, A. Khenchaf, V. Guardia, Evaluation of a two-step clustering method for radar pulse deinterleaving, in: 2023 IEEE Conference on Antenna Measurements and Applications (CAMA), IEEE, 2023, pp. 1047–1051.
[13] M.A. Nuhoglu, H.A. Cirpan, Radar signal deinterleaving in electronic warfare systems: a combined approach, IEEE Access 11 (2023) 142043–142061.
[14] M. Mottier, G. Chardon, F. Pascal, Deinterleaving and clustering unknown radar pulses, in: 2021 IEEE Radar Conference (RadarConf21), IEEE, 2021, pp. 1–6.
[15] S. Zhou, Z. Xu, F. Liu, Method for determining the optimal number of clusters based on agglomerative hierarchical clustering, IEEE Trans. Neural Netw. Learn. Syst. 28 (2016) 3007–3017.
[16] K. Kang, Y.-x. Zhang, W.-p. Guo, L.-g. Tian, Key radar signal sorting and recognition method based on clustering combined with pri transform algorithm, J. Artif. Intell. Technol. 2 (2022) 62–68.
[17] Z. Xiao, Z. Yan, Radar emitter identification based on naive Bayesian algorithm, in: 2020 IEEE 5th Information Technology and Mechatronics Engineering Conference (ITOEC), IEEE, 2020, pp. 1242–1244.
[18] J. Pan, J. Xiong, Y. Zhuo, Working mode recognition of non-specific radar based on resnet-svm learning framework, Sensors 23 (2023) 3123.
[19] L. Cain, J. Clark, E. Pauls, B. Ausdenmoore, R. Clouse, T. Josue, Convolutional neural networks for radar emitter classification, in: 2018 IEEE 8th Annual Computing and Communication Workshop and Conference (CCWC), IEEE, 2018, pp. 79–83.
[20] L. Wang, S. Mi, Y. Tian, F. Wang, S. He, The polar centernet for radar signal identification, in: 2022 4th International Conference on Intelligent Information Processing (IIP), IEEE, 2022, pp. 14–17.
[21] D.H. Park, D.-H. Seo, J.-H. Baek, W.-J. Lee, D.E. Chang, A novel batch streaming pipeline for radar emitter classification, Appl. Sci. 13 (2023) 12395.
[22] Z. Liu, Y. Shi, Y. Zeng, Y. Gong, Radar emitter signal detection with convolutional neural network, in: 2019 IEEE 11th International Conference on Advanced Infocomm Technology (ICAIT), IEEE, 2019, pp. 48–51.
[23] K. Li, Radar emitter identification based on improved convolutional neural network, in: 2019 IEEE 3rd Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC), IEEE, 2019, pp. 118–121.
[24] Z. Zhou, G. Huang, H. Chen, J. Gao, Automatic radar waveform recognition based on deep convolutional denoising auto-encoders, Circuits Syst. Signal Process. 37 (2018) 4034–4048.
[25] K. Chen, S. Zhang, L. Zhu, S. Chen, H. Zhao, Modulation recognition of radar signals based on adaptive singular value reconstruction and deep residual learning, Sensors 21 (2021) 449.
[26] E. Yar, M.B. Kocamis, A. Orduyilmaz, M. Serin, M. Efe, A complete framework of radar pulse detection and modulation classification for cognitive ew, in: 2019 27th European Signal Processing Conference (EUSIPCO), IEEE, 2019, pp. 1–5.
[27] M. Zhang, M. Diao, L. Guo, Convolutional neural networks for automatic cognitive radio waveform recognition, IEEE Access 5 (2017) 11074–11082.
[28] J. Zhang, Y. Li, J. Yin, Modulation classification method for frequency modulation signals based on the time–frequency distribution and cnn, IET Radar Sonar Navig. 12 (2018) 244–249.
[29] Z. Huang, Z. Ma, G. Huang, Radar waveform recognition based on multiple autocorrelation images, IEEE Access 7 (2019) 98653–98668.
[30] S. Wei, Q. Qu, H. Su, J. Shi, X. Zeng, X. Hao, Intra-pulse modulation radar signal recognition based on squeeze-and-excitation networks, Signal Image Video Process. 14 (2020) 1133–1141.
[31] M. Zhang, M. Diao, L. Gao, L. Liu, Neural networks for radar waveform recognition, Symmetry 9 (2017) 75.
[32] H. Mardia, New techniques for the deinterleaving of repetitive sequences, in: IEE Proceedings F, in: Radar and Signal Processing, vol. 136, IET, 1989, pp. 149–154.
[33] D. Milojević, B.M. Popović, Improved algorithm for the deinterleaving of radar pulses, in: IEE Proceedings F, in: Radar and Signal Processing, vol. 139, IET, 1992, pp. 98–104.
[34] T. Tian, Q. Zhang, Z. Zhang, F. Niu, X. Guo, F. Zhou, Shipborne multi-function radar working mode recognition based on dp-atcn, Remote Sens. 15 (2023) 3415.
[35] U.I. Ahmed, T. ur Rehman, S. Baqar, I. Hussain, M. Adnan, Robust pulse repetition interval (pri) classification scheme under complex multi emitter scenario, in: 2018 22nd International Microwave and Radar Conference (MIKON), IEEE, 2018, pp. 597–600.
[36] S.Q. Wang, J. Bai, X.Y. Huang, C.Y. Gao, P.F. Wan, Analysis of radar emitter signal sorting and recognition model structure, Proc. Comput. Sci. 154 (2019) 500–503.
[37] W. Qu, G. Yao, L. Meng, Research on radar pri modulation pattern recognition based on recurrent neural network, in: 2023 4th International Conference on Computer Vision, Image and Deep Learning (CVIDL), IEEE, 2023, pp. 250–254.
[38] Q. Wang, P. Du, J. Yang, G. Wang, J. Lei, C. Hou, Transferred deep learning based waveform recognition for cognitive passive radar, Signal Process. 155 (2019) 259–267.
[39] N. Daldal, Ö. Yıldırım, K. Polat, Deep long short-term memory networks-based automatic recognition of six different digital modulation types under varying noise conditions, Neural Comput. Appl. 31 (2019) 1967–1981.
[40] A.A. Haque, K. Xie, K. Liu, H. Zhao, X. Yang, X. Yuan, Detection of copy number variations from ngs data by using an adaptive kernel density estimation-based outlier factor, Digit. Signal Process. 126 (2022) 103524.
[41] M. Tan, Q. Le, Efficientnet: Rethinking Model Scaling for Convolutional Neural Networks, 2019, pp. 6105–6114.
[42] D. Steinley, Properties of the Hubert-Arable adjusted rand index, Psychol. Methods 9 (2004) 386.
[43] S. Xu, L. Liu, Z. Zhao, Unsupervised recognition of radar signals combining multi-block tfr with subspace clustering, Digit. Signal Process. 151 (2024) 104552.

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